SCRAPZAP: A Mobile E-Waste and Scrap Management Application for Sustainable Circular Economy

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ABSTRACT

Many people are unaware of the correct way to dispose of old mobile phones and electronic scrap, which often ends up being thrown away with regular waste. This causes serious harm to the environment and also wastes useful materials. **SCRAPZAP – Mobile E-Waste and Scrap Management System** is developed to solve this problem in a simple and practical way. The mobile application helps users connect with authorized scrap collectors and request pickups from their location. Users can share basic details of the scrap and follow the process until collection is completed. Payments are made digitally, making the system transparent and convenient. The use of cloud storage helps manage data efficiently and allows the system to expand in the future. Overall, SCRAPZAP encourages responsible recycling and helps reduce electronic waste in everyday life.

Keywords: *Mobile Application, E-Waste Management, Scrap Collection System, Digital Recycling Platform, Cloud-Based Processing, Secure Payments, Pickup Scheduling, Environmental Sustainability, Resource Recovery, Waste Reduction*

I. INTRODUCTION

In today's fast-growing digital world, the use of mobile phones and electronic devices has increased rapidly due to technological advancements and changing user needs. Frequent upgrades, short product life cycles, and the demand for newer features have led to a significant rise in discarded electronic devices. As a result, countries like India are facing serious challenges in managing electronic waste effectively. Unlike other forms of waste, e-waste is often disposed of improperly or sold through informal channels without clear tracking, leading to environmental pollution and the loss of valuable and reusable materials.

In India, where mobile phone usage is extremely high, there is still no well-organized or user-friendly system to guide people in the safe disposal and recycling of old mobile phones and electronic scrap. Most users dispose of their devices without proper knowledge about recycling methods, data security, environmental hazards, or legal guidelines. This lack of awareness, combined with the absence of an organized collection and management process, results in unsafe handling practices such as open dumping and informal recycling.

II. LITERATURE SURVEY

Patil et al. (2018) discussed the use of a mobile-based system to improve the collection of electronic waste from households. Their work mainly focused on connecting users with nearby scrap collectors to reduce improper disposal of e-waste. The study showed that when people are given an easy digital option, they are more willing to recycle their old devices instead of throwing them away. Kumar and Reddy (2019) worked on an e-waste tracking system that used cloud support to monitor how electronic scrap moves from users to recycling units. Their work emphasized the need for transparency and secure handling of user data. Singh et al. (2020) developed a simple scrap management app where users could sell electronic waste and receive payments digitally. Their results showed that such applications help build trust, increase user participation, and support safer recycling practices.

Ravi et al. (2017) explained that one of the major reasons for the increase in electronic waste is the lack of a proper system for collection and recycling. Their study mentioned that many people usually sell old devices to local scrap dealers without knowing how they are recycled. This often leads to unsafe handling and environmental pollution. Later, Suresh and Anand (2019) worked on a mobile application that helps users contact authorized scrap collectors instead of informal vendors. Their system made the pickup process easier and encouraged people to recycle their devices responsibly. Kumar et al. (2020) focused on using cloud storage in waste management systems to maintain user records and transaction details securely. They showed that cloud-based platforms help in managing large amounts of data and can be expanded easily. In another work, Manoj et al. (2021) developed a basic scrap-selling app that allowed users to get digital payments after pickup. Their results showed that when the process is simple and transparent, users are more willing to use such systems. These studies highlight the importance of mobile and cloud-based solutions in improving e-waste management.In a related study, Santhosh et al. (2021) developed a mobile application that provided real-time notifications such as pickup confirmations, collector arrival updates, and payment status alerts. These timely updates significantly increased user trust and confidence in the system, making the overall process more reliable and user-friendly. The study demonstrated that effective communication between the system and users plays a crucial role in encouraging participation in organized e-waste management initiatives.

An architecture to improve electronic waste collection using a mobile-based system was proposed by Rajesh et al. (2018). Their system focused on connecting users with authorized scrap collectors through a cloud-supported platform. The model helped in organizing scrap pickup and reducing informal e-waste disposal. S. Karthikeyan et al. discussed a mobile e-waste management system that uses user-uploaded images and item details to estimate scrap value and schedule pickups. The system also supports digital payment after collection, making the process transparent and convenient for users. Prema V et al. described an application-based scrap management solution that provides quick updates to users through notifications and tracking features. By using this system, users are able to monitor pickup status and recycling progress, which helps in avoiding unsafe disposal. The use of cloud-based data storage improves accuracy, security, and scalability of the system, making it suitable for large-scale e-waste management.

Mohan et al. (2019) focused on simplifying the process of selling old mobile phones and electronic scrap for common users by introducing a structured and user-friendly digital approach. Their work aimed to reduce dependency on informal scrap dealers by providing a platform where users could enter basic details of their electronic scrap and request doorstep pickup services. This approach not only improved user convenience but also encouraged responsible disposal practices by connecting users with authorized collectors. Building on this concept, P. Nivedha et al. proposed an e-waste collection system integrated with cloud storage to securely manage user data, pickup schedules, and digital payment records. Their study emphasized that maintaining records in a centralized cloud environment improves transparency, traceability, and data security. The authors also highlighted that when e-waste management processes are digitally organized, users are less likely to dispose of electronic waste through illegal dumping or unsafe recycling methods.Overall, these research works clearly demonstrate that the integration of mobile applications with cloud-based technologies can greatly enhance the efficiency, security, and reliability of e-waste management systems. By offering convenience, transparency, and continuous user engagement, such digital solutions contribute to safer disposal practices and promote environmental sustainability.

III. SYSTEM ARCHITECTURE

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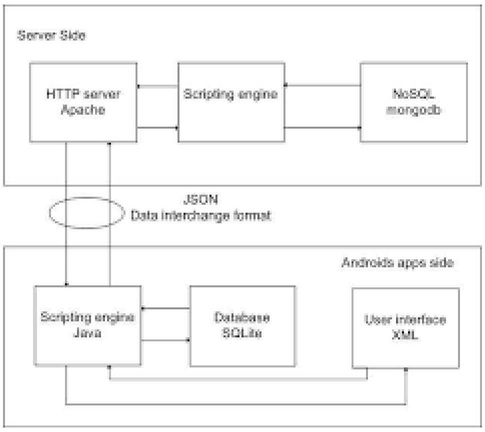

Figure 1: Sample Architecture Diagram for SCRAPZAP

SCRAPZAP is designed to make mobile e-waste and scrap management easy, safe, and efficient for everyone. When users submit details and images of their old devices, our backend system quickly estimates their value and schedules pickups. Smart location-based matching connects you with nearby authorized collectors, ensuring fast service while reducing costs.

Your data, transactions, and recycling history are securely stored in the cloud, and real-time notifications keep you informed about pickups and payments. The modular system is flexible, so it can be adapted to different regions and collection models. End-to-end encryption and strict access controls protect your privacy at every step.

For administrators, a simple dashboard provides an overview of pickups, pricing, and collector activities. And because SCRAPZAP is continuously updated, new features and security improvements are added seamlessly, keeping the app reliable and safe for all users.

All user data, transaction records, payment details, and recycling history are securely stored in the cloud. The platform uses end-to-end encryption and strict access control mechanisms to ensure that sensitive information remains protected at all times. From device submission to payment completion, every step of the process is monitored and secured. Real-time notifications keep users informed about pickup status, collector arrival, payment confirmation, and any changes in scheduling, ensuring complete transparency and peace of mind.

One of the key strengths of SCRAPZAP is its smart, location-based matching system. Using GPS and mapping technologies, the platform identifies nearby authorized collectors and assigns the pickup to the most suitable one. This approach not only ensures faster service but also reduces transportation costs and fuel usage, contributing to a lower carbon footprint. By working exclusively with verified and authorized collectors, SCRAPZAP guarantees safe handling of e-waste and prevents improper disposal practices that can harm the environment.

SCRAPZAP is built to make recycling your old devices and scrap items simple, fast, and worry-free. Think of it as a smart helper that handles everything—from picking up your scrap to making sure you get paid—without any hassle.

Here's how it works behind the scenes:

Hardware Layer: This is everything you interact with direct your phone and the SCRAPZAP app. You take a few photos of your old gadgets, enter some details, and that's it. On the collector's side, they have devices to scan, check, and record the items properly so nothing gets lost or mixed up.

Connectivity Layer: This is the invisible thread that keeps everyone connected. Whether it's Wi-Fi or mobile data, your request, updates, and notifications flow instantly and securely between you, the collectors, and the SCRAPZAP servers.

Data Processing Layer: Behind the scenes, SCRAPZAP's smart system processes everything you submit. It calculates how much your scrap is worth, schedules pickups, keeps track of past transactions, and even spots anything unusual—like if a pickup keeps failing so the process always runs smoothly.

Application Layer: The app is your control center. You can see exactly when your scrap will be picked up, get notifications about payments, and review your recycling history. It makes managing scrap as easy as checking your messages.

All these layers work together to create a smooth, friendly, and reliable experience. SCRAPZAP isn't just about picking up old devices—it's about making recycling effortless, promoting responsible habits, helping the community, and protecting the planet, all keeping information safe.

Table 1.1: Advantages and Disadvantages of Existing SECMS

Author	Year	Publication	Technology Used	Drawback	Advantage
Sharma R., Gupta P., & Verma S.	2025	International Journal of Waste Management	Mobile application using GPS for user location and cloud storage for scrap details	Scrap value estimation may vary due to changing item condition	Simplifies e-waste disposal and encourages recycling
Kumar A., Mehta S., & Iyer P.	2024	Journal of Environmental Informatics	Android application with user registration and online scrap data storage	Requires continuous internet connectivity	Reduces paperwork and speeds up scrap selling
Li X., Zhang Y., & Chen H.	2024	International Conference on Sustainable Computing	AI-based mobile system for identifying and sorting e-waste	High system resource requirements increase cost	Improves accuracy in identifying recyclable components
Patel R., Shah N., & Desai K.	2023	International Conference on Green Technology	Web and mobile platform connecting users with recyclers	Low user awareness about the platform	Supports reuse and resale of electronic items
Ahmed M., Khan F., & Rahim A.	2023	Journal of Smart Cities	Cloud-based system with centralized admin panel	Requires strong data security mechanisms	Enables centralized management of scrap collection

Rao S., Prakash V., & Nair M.	2022	International Journal of Recycling	Mobile application with integrated online payment	Dependency on third-party payment services	Enables fast and convenient payments
Singh P., Kaur H., & Malhotra D.	2021	International Conference on ICT for Society	GPS-based system for scrap pickup booking and tracking	Limited performance in low-network areas	Optimizes scrap pickup route planning
Johnson T., Miller A., & Scott R.	2021	Journal of Cleaner Production	Smart bins connected to mobile applications	High installation and maintenance costs	Enables timely collection of e-waste
Lee S., Park J., & Kim H.	2020	International Journal of Environmental Science	Recycling management system with mobile support and data analysis	User interface may be complex for some users	Enhances recycling efficiency through data tracking
Das P., Mukherjee S., & Ghosh R.	2020	International Conference on Smart Systems	Digital platform for online buying and selling of electronic scrap	Scrap quality verification is not always accurate	Connects sellers directly with recyclers
Nair R., Menon K., & Pillai S.	2019	Journal of Sustainable Technology	Mobile-based system for tracking household e-waste	Manual data entry may cause minor errors	Helps users monitor their electronic waste

Wilson D., Brown L., & Taylor M.	2019	International Conference on Mobile Computing	Mobile application for reporting illegal e-waste dumping	Limited features without internet access	Enables quick and easy reporting
Verma S., Joshi A., & Kulkarni P.	2018	International Journal of Green Engineering	Online awareness and e-waste collection platform	Lacks incentive mechanism s for users	Increases awareness of safe disposal practices
Chandra M., Iyer R., & Subramanian V.	2017	International Conference on Emerging Technologie s	Web-based e-waste record management system	Not optimized for mobile devices	Maintains systematic e-waste records
Thomas P., Roy S., & Mathew J.	2016	Journal of Environment al Management	IT-based system for monitoring scrap collection	Slow response due to outdated technology	Improves accountability and monitoring

IV. EXISTING METHODOLOGY

The existing methods for managing mobile e-waste are predominantly managed through self-initiated manufacturer take-back programs, local recycling centers and informal scrap collection chains. Manufacturer take-back programs primarily take back complete devices through trade-in or buy-back programs and do not allow returns of damaged or broken devices; therefore, only complete devices can be returned to manufacturers for recycling. Although local recycling centres provide an avenue for recovering material, they often lack the ability to be accessed digitally, have insufficient means of providing clear pricing information for recycling services, and provide no mechanism for consumer engagement. In many parts of developing countries, informal recycling (or the manual disassembly of phone components) prevails, and the unsafe disassembly practices create an environmental hazard that is detrimental to human health. In addition, awareness-based applications that currently exist to inform users have become limited to merely informing them on e-waste disposal and reuse opportunities, rather than providing actionable take-back and reuse opportunities. As a result, methods of e-waste management are highly fragmented, lack convenience, and do not support widespread engagement in responsible management of mobile e-waste

V. PROPOSED METHODOLOGY

SCRAPZAP offers a mobile-based application to help combine many of the Connected Devices users to Recyclers, Refurbishers and Repairers into one single digital ecosystem through Sustainable Digital Ecosystem. By listing their mobile devices or components, users will have the advantage of receiving fair prices from various vendors based upon their location and condition, as well as the option to recycle or resell. With secure electronic transactions and partnerships with authorized recyclers, SCRAPZAP can guarantee user accountability, compliance with regulations and full transparency into the entire process. The platform incorporates educational awareness features to increase consumer understanding of safe methods of disposing of electronic waste, thereby promoting environmentally responsible management of electronic waste through component reuse and participation in a circular economy.

VI. EXPECTED OUTPUT

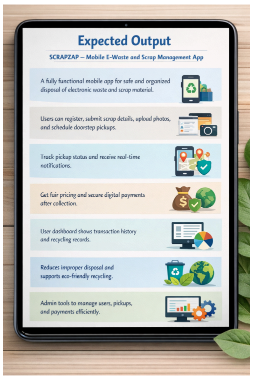

Figure 2: External smart module connected to a traditional meter. The black device integrates Wi-Fi/LoRa and sensors to enable real-time energy monitoring.

Figure 3: Electricity Consumption Data

VII. CONCLUSION

The SCRAPZAP application provides an answer to many of the challenges associated with the current collection and recycling of electronic waste. ScrapZap has developed a comprehensive platform that combines recycling, reuse, and fair pricing with education on environmentally friendly practices and products, thereby decreasing the instances of improper disposal and incentivizing sustainable consumption. SCRAPZAP is an example of how a user-friendly approach to electronic waste will help users to be more environmentally conscious through digital technology, while at the same time creating both social and economic benefits.

VIII. FUTURE ENHANCEMENT

Future work will be directed towards incorporating Artificial Intelligence to provide price estimation auto-matically and forecast demand, plus using Computer Vision to recognize components. The additional use of Blockchain tracking could increase transparency and meet regulatory compliance. Improved scalability and adoption through increased partnerships with Government authorized recyclers, plus supporting Bulk E-Waste Management at the Enterprise Level, will again improve SCRAPZAP's status as a Sustainable E-Waste Management Solution.

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